

Project — Fit a Linear Model to Neural Responses

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given a table of trials for one neuron. Each trial has a stimulus (orientation and contrast) and the neuron's response. Your job is to fit a linear model that predicts the response from the stimulus, and check how well it works.

Background (short)

A General Linear Model (GLM) says:

$$\text{response} = \beta_0 + \beta_1 \cdot \text{orientation} + \beta_2 \cdot \text{contrast} + \text{noise}$$

Fitting the model means finding the β values that best explain the data. This is ordinary least squares (linear regression).

Data

In `shared/data/`:

- `trials.csv` — one row per trial with columns for stimulus features and the measured response.
- `data_dictionary.csv` — what each column means.

Steps

1. Load and look. Open `student/notebooks/00_starter.ipynb`. Load `trials.csv` with `pandas.read_csv`. Print the first 5 rows. Plot the response distribution (a histogram).
2. Build the design matrix and fit. Choose orientation and contrast as your predictors. Fit with `sklearn.linear_model.LinearRegression` or `statsmodels.OLS`. Report the fitted β_0 , β_1 , β_2 .
3. Check the fit. Plot the predicted response vs. the observed response (a scatter plot, one dot per trial). A good fit lines up on the diagonal. Report the R^2 .
4. Simple validation. Split the data 80/20 (train/test) with `train_test_split`. Fit on train, evaluate R^2 on test. Compare with the R^2 from Step 3.

What to hand in

- Your filled notebook.
- The predicted-vs-observed scatter plot.
- Fitted coefficients (β_0 , β_1 , β_2) and both R^2 values (train and test).
- A short paragraph (3-4 sentences): which stimulus feature drives the response more? Does the model generalise?

If you finish early: add an interaction term (orientation * contrast) and see if the test R^2 improves.